

**Project Interim Report
On**

Livestock Breed ID

**(Using image classification to identify cattle breeds
for government digital census records)**

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1. Abstract

The accurate identification of cattle breeds is a crucial component of modern livestock management and precision agriculture. This project presents an end-to-end Machine Learning web application designed to classify various Indian cattle breeds from images. To ensure high robustness, the system employs a two-step pipeline: an initial object detection phase using YOLOv8 to locate and crop cattle from complex backgrounds, followed by a classification phase using Transfer Learning with MobileNetV2. The solution features a decoupled architecture, incorporating a centralized ML Core, a high-performance FastAPI backend, and a modern React-based frontend to provide an intuitive user experience for single and multi-image breed prediction.

2. Introduction

2.1 Problem Statement

In the agricultural sector, identifying cattle breeds accurately is essential for breeding programs, health monitoring, and market valuation. Manual identification relies heavily on human expertise, which can be subjective, time-consuming, and error-prone. Additionally, field images often contain background noise, multiple animals, or irrelevant objects. There is a need for an automated, reliable, and accessible tool that can accurately isolate and classify cattle breeds using computer vision.

2.2 Objectives

- **Automate Breed Identification:** Eliminate the need for expert-level taxonomic knowledge in the field.
 - **Standardize Census Records:** Ensure that the data collected across different regions follows a uniform classification standard.
 - **Reduce Manual Entry Work:** Automate the logging of breed data into digital databases to prevent transcription errors.
 - **Facilitate Digital Integration:** Provide an API-ready structure that can be integrated into existing government census mobile applications.
-

3. Scope

This project is specifically designed for the classification of major indigenous and exotic cattle breeds. While currently focused on cattle, the modular architecture allows for future extension to other livestock species such as buffaloes, goats, and sheep. It is intended for use by government field officers and veterinary departments.

4. System Requirements

4.1 Hardware Requirements

- * **Processor:** Minimum Intel Core i5 or equivalent (i7/i9 or AMD Ryzen 5+ recommended for faster inference).
- * **RAM:** Minimum 8 GB (16 GB or higher recommended, especially for model training).
- * **Storage:** 5 GB of free disk space for models, datasets, and project files.
- * **GPU (Optional but Recommended):** NVIDIA GPU with CUDA support for accelerated model training and inference.

4.2 Software Requirements

- * **Operating System:** Windows 10/11, macOS, or Linux.
 - * **Programming Language:** Python 3.8+
 - * **Frontend Environment:** Node.js 18+ and npm.
 - * **Libraries & Frameworks:**
 - * Backend: FastAPI, Uvicorn, Python-Multipart.
 - * Machine Learning: TensorFlow, Keras, Ultralytics YOLO, OpenCV, NumPy.
 - * Frontend: React, Vite.
-

5. Methodology

The methodology involves a robust two-step pipeline:

5.1: Object Detection with YOLOv8

Images uploaded to the system are first processed by a YOLOv8s (You Only Look Once) model.

- * The model is configured with a high sensitivity (e.g., confidence threshold of 0.20) to detect cattle (COCO class ID 19).
- * Upon detection, the bounding box coordinates are extracted. A 10% padding is added to the bounding box to ensure the entire animal is captured.
- * The image is cropped to this bounding box, effectively removing background noise and isolating the subject.
- * If no cattle are detected, the system intelligently aborts the classification phase and returns a "No Cattle Detected" response to the user.

5.2: Image Preprocessing and Classification

The cropped cattle image is then processed for classification:

- * **Preprocessing:** The cropped image is resized to standard dimensions (224x224 pixels) and normalized.
- * **Transfer Learning with MobileNetV2:** MobileNetV2, pre-trained on ImageNet, is used as the base feature extractor. Its lightweight architecture provides an optimal balance of speed and accuracy.
- * **Custom Classification Head:** The top classification layers are replaced with a custom Dense neural network tailored to output probabilities for the specific cattle breeds in the dataset.

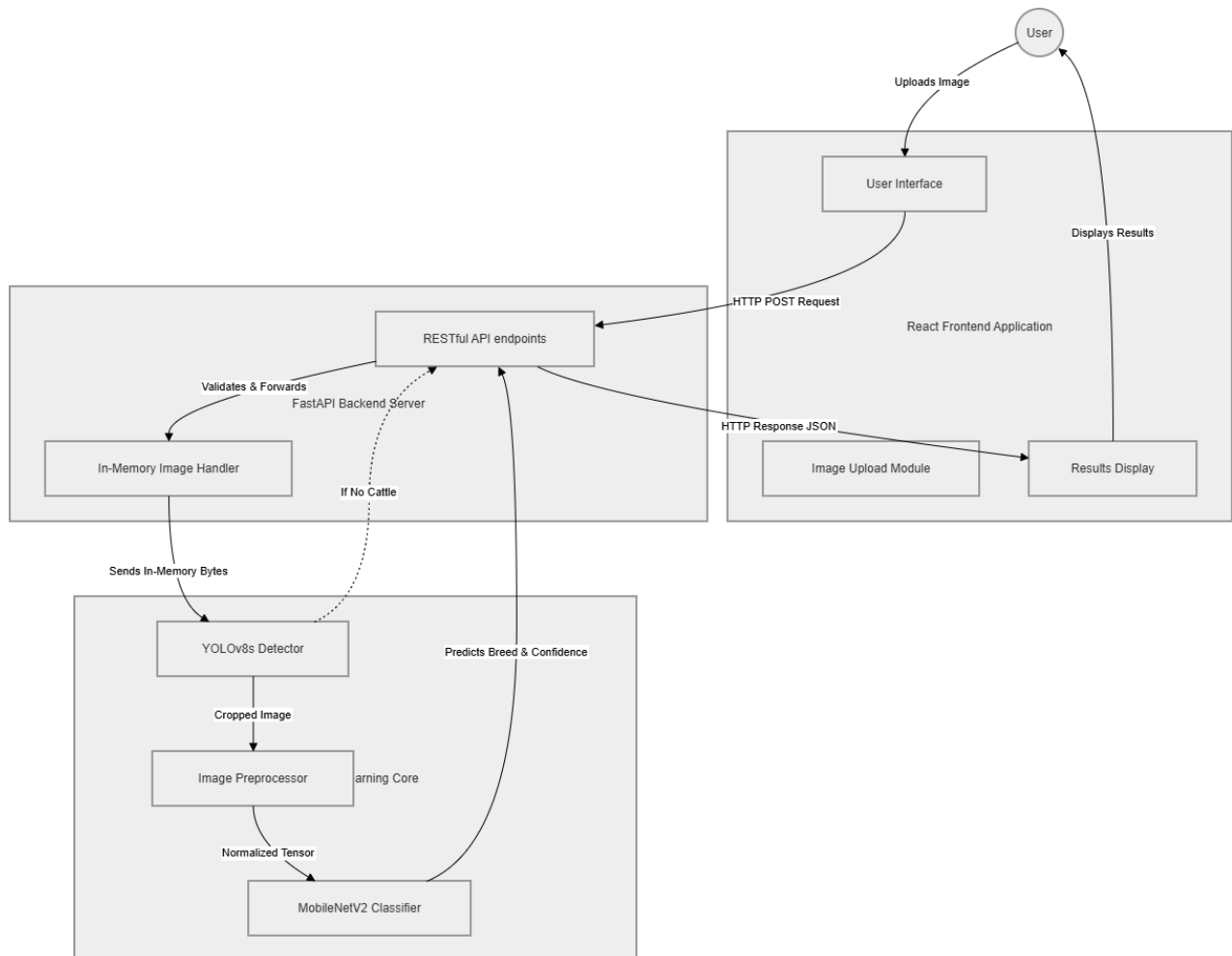
5.3 Dataset Description

The project utilizes the **Indian Cattle Image Dataset** (sourced from Kaggle). This dataset comprises categorized folders containing images of various distinct Indian cattle breeds (such as Amritmahal, Gir, etc.). The dataset is organized in a hierarchical directory structure where each subdirectory represents a specific breed class.

6. System Design

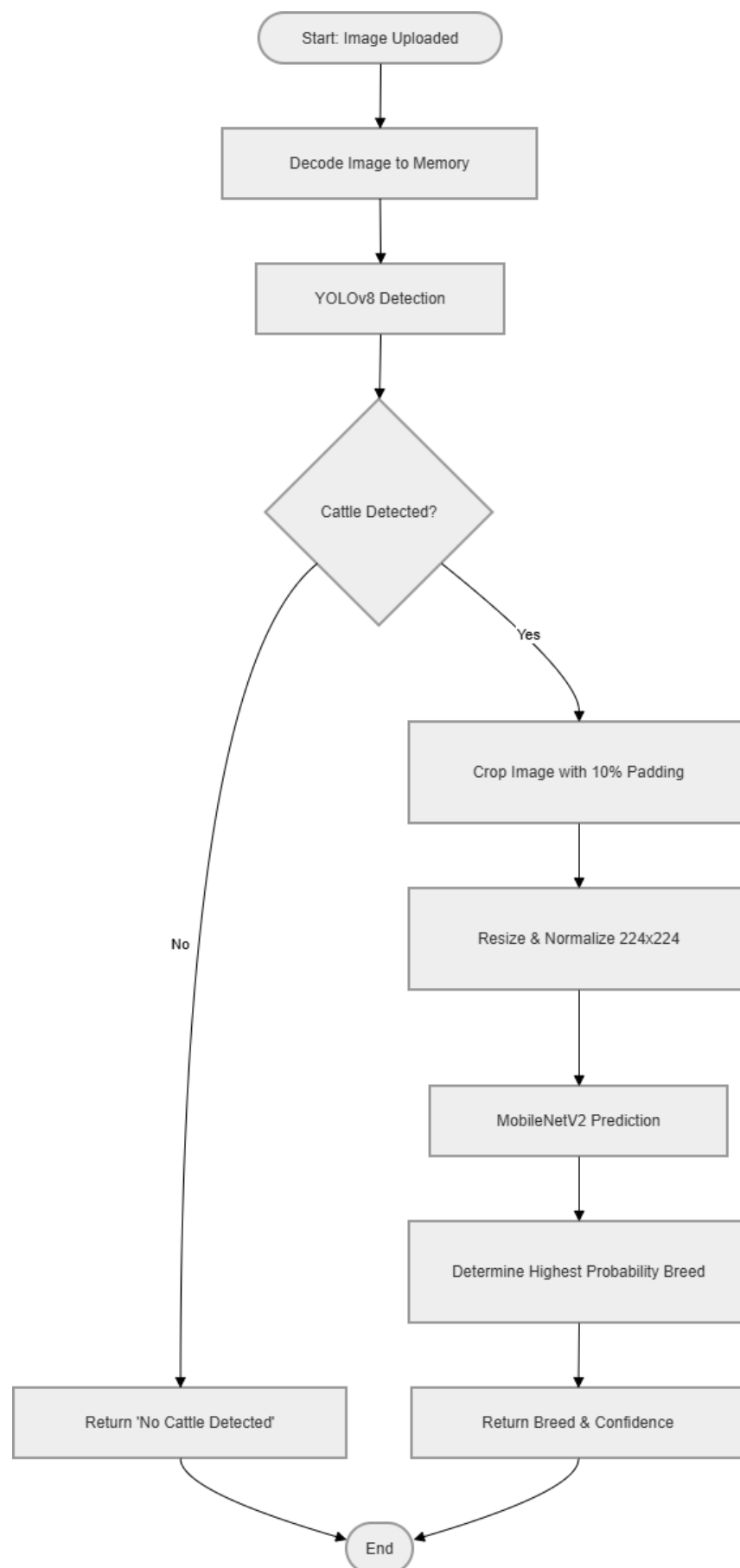
6.1 Architecture Diagram

The system follows a client-server architecture, distinctly separating presentation, API routing, and the machine learning inference engine.



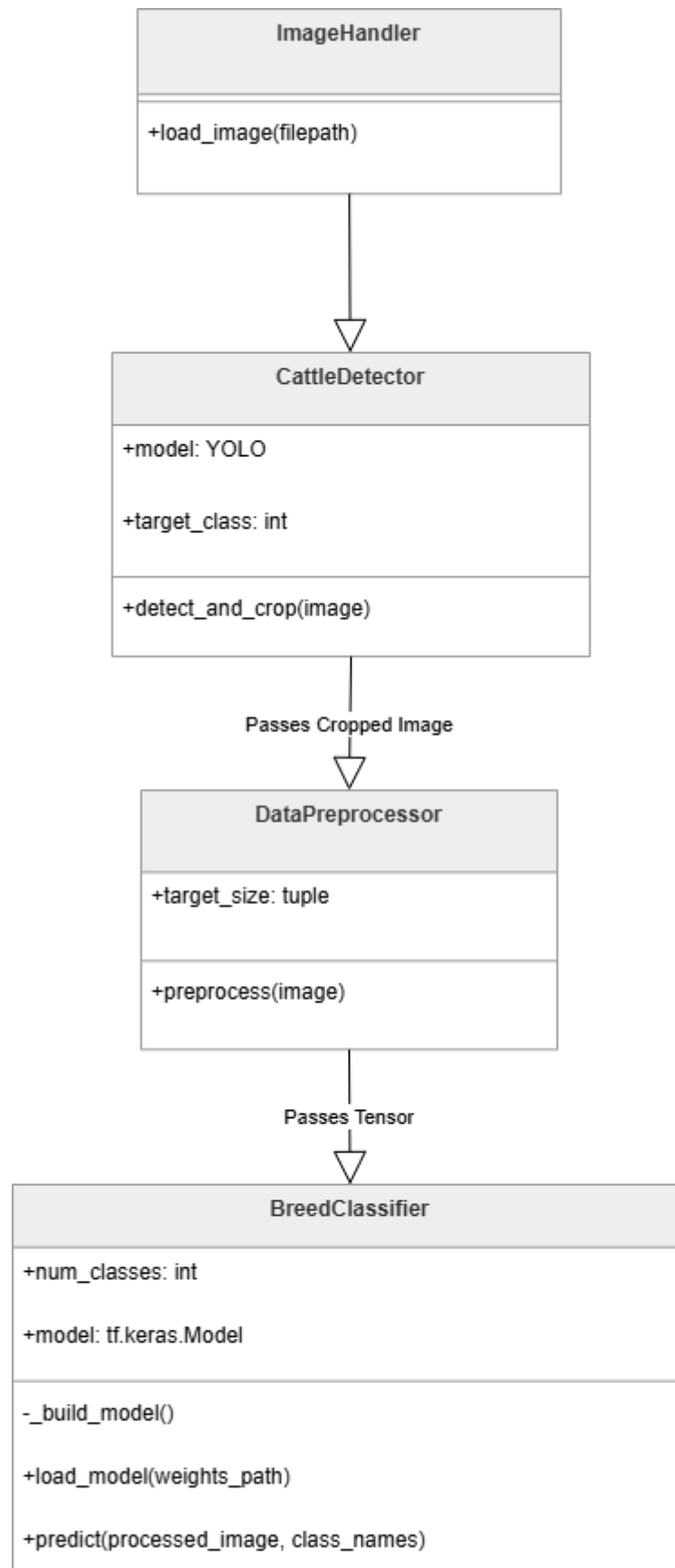
6.2 Flowchart

The following flowchart illustrates the image processing pipeline during inference:



6.3 Class Diagram

The core machine learning logic in `ml\_core.py` is structured using Object-Oriented Programming principles.



7. Implementation

7.1. Logic of Classes

- **CattleDetector:** Acts as the initial "gatekeeper" using a YOLOv8 object detection model. It identifies if a cow is present in the image and crops it with a 10% padding to ensure the whole animal is visible for classification. If no cow is detected, the process stops.
- **DataPreprocessor:** Handles the conversion of raw cropped images into a mathematical format suitable for the neural network. It resizes the images to 224x224 pixels and normalizes pixel values to a 0–1 scale.
- **BreedClassifier:** The core classification engine that utilizes the MobileNetV2 architecture. It loads the pre-trained weights, defines the custom neural network head, and performs the forward pass to determine the specific cattle breed and confidence level.
- **ImageHandler:** Manages I/O operations by ensuring that images are correctly read from a file path and converted from BGR (OpenCV default) to RGB format for consistent processing.

8. Sample Input/Output

- **Input:**
A user uploads a high-resolution JPG image of a cattle specimen through the React frontend or CLI.
- **Processing:**
The system first runs the YOLOv8 detector to find the cow, crops the image, and then the BreedClassifier loads the livestock_mobilenetv2.weights.h5 to perform the prediction.
- **Output:**

File: image_142x107_65.jpg

Result: **Kenkatha**

Confidence: 76.91%

9. Future Enhancements

- * **Dataset Augmentation:** Incorporating more diverse images across different lighting conditions and angles to further improve model robustness.
- * **Mobile Application:** Porting the React frontend to React Native to provide a dedicated mobile application for on-the-field use.
- * **Cloud Deployment:** Containerizing the application using Docker and deploying it to cloud platforms (e.g., AWS, GCP).

10. Planning & Structure

10.1. Gantt Chart



10.2 Time Management

The use of a Gantt chart allowed for precise tracking of milestones, helping to maintain momentum and avoid last-minute integration issues. Key focus areas included:

- **Start and End Dates:** Defining exact windows for training the BreedClassifier on the Indian Cattle Image Dataset.
 - **Task Dependency:** Ensuring that the FastAPI backend logic was fully tested with the `Testing_model.py` CLI script before deploying the React frontend.
 - **Adaptive Buffers:** Allocating extra time for "Two-Step" refinement to ensure that non-cattle images (like humans or other animals) were correctly identified and filtered.
-

11. Conclusion

The Cattle Breed Classification System successfully demonstrates a robust application of computer vision in precision agriculture. By combining the powerful object detection capabilities of YOLOv8 with the efficient feature extraction of MobileNetV2, the system handles real-world images effectively. The modern web stack (React and FastAPI) provides a fast and accessible interface, making it a valuable tool for livestock identification and management.

12. References

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6. FastAPI Documentation. Available: <https://fastapi.tiangolo.com/>
7. React Documentation. Available: <https://react.dev/>
8. Darpude, A. "Indian Cattle Image Dataset." Kaggle. Available: <https://www.kaggle.com/datasets/atharvadarpu/indian-cattle-image-dataset>

13. Project Directory Structure and Source Code

Below is the directory layout of the complete Cattle Breed Classification System project:

ML-Project/

```
|— Back_End/
|   |— app.py          # FastAPI web server
|   |— yolov8s.pt      # YOLOv8 weights (cattle detector)
|— Front_End/
|   |— index.html      # HTML Entry point
|   |— vite.config.js  # Vite configuration
|   |— package.json    # Frontend dependencies
|   |— src/
|       |— main.jsx    # React entry script
|       |— App.jsx     # Main React App UI
|       |— index.css   # Custom styling
|— cattle/              # Dataset folder structured by breed names
|— ml_core.py          # Central machine learning engine classes
|— Training_model.py   # Script to train classifier & perform inference test
|— Testing_model.py    # Interactive CLI testing script
|— Project_Report.md   # Project report documentation (This file)
|— README.md          # Project setup & run documentation
|— livestock_mobilenetv2.weights.h5 # MobileNetV2 trained classifier weights
```

13.1 ml_core.py

```
import os
import cv2
import numpy as np
import tensorflow as tf
from ultralytics import YOLO

# Configuration constants
DATASET_DIR = 'cattle'
WEIGHTS_PATH = 'livestock_mobilenetv2.weights.h5'
IMG_SIZE = (224, 224)
BATCH_SIZE = 32

def get_target_breeds(dataset_dir=DATASET_DIR):
    """Dynamically get class names based on folder structure."""
    if os.path.exists(dataset_dir):
        return sorted([d for d in os.listdir(dataset_dir) if os.path.isdir(os.path.join(dataset_dir, d))])
    return []

class CattleDetector:
    """Step 1: Detects ONLY cattle using YOLOv8 with high sensitivity."""
```

```

def __init__(self, model_name='yolov8s.pt'):
    # Using yolov8s.pt for better accuracy than the 'nano' version
    self.model = YOLO(model_name)
    # COCO ID 19 is strictly for 'cow'
    self.target_class = 19

def detect_and_crop(self, image):
    """Returns a cropped image of the cow if found, otherwise None."""
    # conf=0.20 makes the detector very sensitive to cows in the background
    results = self.model(image, conf=0.20, verbose=False)

    for result in results:
        for box in result.bboxes:
            # Check ONLY for the cow class
            if int(box.cls[0]) == self.target_class:
                x1, y1, x2, y2 = map(int, box.xyxy[0])

                # Add 10% padding to ensure the whole animal is visible for the classifier
                h, w, _ = image.shape
                pad_w = int((x2 - x1) * 0.1)
                pad_h = int((y2 - y1) * 0.1)

                x1_new = max(0, x1 - pad_w)
                y1_new = max(0, y1 - pad_h)
                x2_new = min(w, x2 + pad_w)
                y2_new = min(h, y2 + pad_h)

                return image[y1_new:y2_new, x1_new:x2_new]
    return None

```

```

class ImageHandler:
    """Manages I/O operations for cattle images."""
    @staticmethod
    def load_image(filepath):
        if not os.path.exists(filepath):
            raise FileNotFoundError(f"Image not found at path: {filepath}")
        image = cv2.imread(filepath)
        if image is None:
            raise ValueError(f"Failed to load image: {filepath}")
        return cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

```

```

class DataPreprocessor:
    """Handles the conversion of raw images into model format."""
    def __init__(self, target_size=IMG_SIZE):
        self.target_size = target_size

```

```

def preprocess(self, image):
    resized_image = cv2.resize(image, self.target_size)
    normalized_image = resized_image.astype(np.float32) / 255.0
    return np.expand_dims(normalized_image, axis=0)

class BreedClassifier:
    """Step 2: The engine for classifying the specific cattle breed."""
    def __init__(self, num_classes, model_weights_path=WEIGHTS_PATH):
        self.num_classes = num_classes
        self.model = self._build_model()
        if model_weights_path and os.path.exists(model_weights_path):
            self.load_model(model_weights_path)

    def _build_model(self):
        """Constructs and compiles the MobileNetV2 architecture."""
        base_model = tf.keras.applications.MobileNetV2(
            weights='imagenet', include_top=False, input_shape=(IMG_SIZE[0], IMG_SIZE[1], 3)
        )
        base_model.trainable = False
        x = tf.keras.layers.GlobalAveragePooling2D()(base_model.output)
        x = tf.keras.layers.Dense(512, activation='relu')(x)
        predictions = tf.keras.layers.Dense(self.num_classes, activation='softmax')(x)
        model = tf.keras.models.Model(inputs=base_model.input, outputs=predictions)
        model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
        return model

    def load_model(self, weights_path):
        self.model.load_weights(weights_path)

    def predict(self, processed_image, class_names):
        """Performs breed classification on the cropped image."""
        predictions = self.model.predict(processed_image, verbose=0)
        idx = np.argmax(predictions[0])
        conf = np.max(predictions[0])
        return class_names[idx], round(float(conf) * 100, 2)

```

13.2 Back_End/app.py

```

import os
import sys
import cv2
import numpy as np
from fastapi import FastAPI, UploadFile, File
from fastapi.middleware.cors import CORSMiddleware

```

```

import uvicorn

# Add the parent directory to the system path to import ml_core
sys.path.append(os.path.abspath(os.path.join(os.path.dirname(__file__), '..')))
from ml_core import (
    DataPreprocessor,
    BreedClassifier,
    CattleDetector,
    get_target_breeds,
    IMG_SIZE,
    WEIGHTS_PATH,
    DATASET_DIR
)

app = FastAPI(title="Cattle Breed Classifier API with Detection")

app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

# Configuration paths
BASE_DIR = os.path.abspath(os.path.join(os.path.dirname(__file__), '..'))
dataset_path = os.path.join(BASE_DIR, DATASET_DIR)
weights_path = os.path.join(BASE_DIR, WEIGHTS_PATH)

TARGET_BREEDS = get_target_breeds(dataset_path)

# Initialize models
try:
    detector = CattleDetector() # Step 1: YOLO Cow Detector
    preprocessor = DataPreprocessor(target_size=IMG_SIZE)
    classifier = BreedClassifier(num_classes=len(TARGET_BREEDS) if TARGET_BREEDS else 1,
    model_weights_path=weights_path)
except Exception as e:
    print(f"Failed to load models: {e}")
    detector, preprocessor, classifier = None, None, None

@app.post("/predict")
async def predict_breeds(files: list[UploadFile] = File(...)):
    if not classifier or not detector:
        return {"error": "Models not properly loaded on server."}

```

```

results = []

for file in files:
    contents = await file.read()
    nparr = np.frombuffer(contents, np.uint8)
    image = cv2.imdecode(nparr, cv2.IMREAD_COLOR)

    if image is None:
        results.append({"filename": file.filename, "error": "Invalid image format"})
        continue

    image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    # 1. Detection Step
    cropped_cow = detector.detect_and_crop(image_rgb)

    if cropped_cow is None:
        # CHANGE: Set breed to specific message and confidence to None
        results.append({
            "filename": file.filename,
            "breed": "No Cattle Detected",
            "confidence": None, # UI will check for this
            "detected": False
        })
        continue

    # 2. Classification Step
    processed_tensor = preprocessor.preprocess(cropped_cow)
    breed, conf = classifier.predict(processed_tensor, TARGET_BREEDS)

    results.append({
        "filename": file.filename,
        "breed": breed,
        "confidence": f"{conf}%",
        "detected": True
    })

return {"results": results}

if __name__ == "__main__":
    uvicorn.run(app, host="0.0.0.0", port=8000)

```

13.3 Front_End/src/App.jsx

```
import { useState } from 'react';
```

```

function App() {
  const [selectedFiles, setSelectedFiles] = useState([]);
  const [predictions, setPredictions] = useState([]);
  const [loading, setLoading] = useState(false);

  const handleFileChange = (e) => {
    setSelectedFiles(Array.from(e.target.files));
  };

  const handleSubmit = async (e) => {
    e.preventDefault();
    if (selectedFiles.length === 0) return alert("Please select at least one image.");

    setLoading(true);
    setPredictions([]);

    const formData = new FormData();
    // Append all selected files to the form data
    selectedFiles.forEach(file => {
      formData.append("files", file);
    });

    try {
      const response = await fetch("http://localhost:8000/predict", {
        method: "POST",
        body: formData,
      });

      if (!response.ok) throw new Error("Network response was not ok");

      const data = await response.json();
      if (data.error) {
        setPredictions([ { filename: 'Server Error', error: data.error } ]);
      } else {
        setPredictions(data.results || []);
      }
    } catch (error) {
      console.error("Error predicting:", error);
      alert("Failed to process images. Is the backend running?");
    } finally {
      setLoading(false);
    }
  };
}

```

```

return (
  <div style={{ maxWidth: '700px', margin: '50px auto', fontFamily: 'sans-serif' }}>
    <h1>Livestock Breed ID System</h1>
    <p>Upload one or multiple cattle images to identify their breed.</p>
    <h1></h1>

    <form onSubmit={handleSubmit} style={{ marginBottom: '20px' }}>
      <input
        type="file"
        multiple
        accept="image/*"
        onChange={handleFileChange}
        style={{ display: 'block', marginBottom: '15px' }}
      />
      <button
        type="submit"
        disabled={loading}
        style={{ padding: '10px 20px', cursor: 'pointer' }}
      >
        {loading ? "Analyzing..." : "Identify Breeds"}
      </button>
    </form>

    {predictions.length > 0 && (
      <div>
        <h2>Results:</h2>
        <ul style={{ listStyle: 'none', padding: 0 }}>
          {predictions.map((pred, index) => (
            <li
              key={index}
              style={{
                border: '1px solid #ccc',
                padding: '15px',
                marginBottom: '10px',
                borderRadius: '8px',
                backgroundColor: pred.confidence ? '#fff' : '#fff9f0' // Light orange tint if no cow
              }}
            >
              <strong>File:</strong> {pred.filename} <br />

              {pred.error ? (
                <span style={{ color: 'red' }}>Error: {pred.error}</span>
              ) : (
                <strong>Result:</strong>

```



```

        <span style={{
            color: pred.confidence ? 'green' : '#d97706',
            fontWeight: 'bold',
            marginLeft: '5px'
        }}>
            {pred.breed}
        </span>
    <br />

    {/* ONLY show confidence if it is NOT null */}
    {pred.confidence && (
        <
            <strong>Confidence:</strong> {pred.confidence}
        </>
    )}
    </>
    )}
</li>
    )}
</ul>
</div>
    )}
</div>
);
}

```

```
export default App;
```

13.4 Training_model.py

```

import os
import tensorflow as tf
from datetime import datetime
from ml_core import ImageHandler, DataPreprocessor, BreedClassifier, get_target_breeds, DATASET_DIR,
WEIGHTS_PATH, BATCH_SIZE, IMG_SIZE

class CensusRecord:
    """Data Transfer Object (DTO) structuring results for the digital census DB."""

    def __init__(self, breed_name, confidence_level):
        self.breed_name = breed_name
        self.confidence_level = round(float(confidence_level) * 100, 2)
        self.timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

    def save_to_db(self):
        """Mocks the database injection process."""

```

```

db_payload = {
    "breed": self.breed_name,
    "confidence": self.confidence_level,
    "recorded_at": self.timestamp
}
success = True # Simulating a successful DB transaction

if success:
    return "Record successfully pushed to Digital Census DB."
else:
    return "Failed to push record to database."

if __name__ == "__main__":
    print("--- Livestock Breed ID System Initializing ---")

    # Pre-check if directory exists
    if not os.path.exists(DATASET_DIR):
        print(f"CRITICAL ERROR: Directory '{DATASET_DIR}' not found.")
        print("Please ensure your dataset folder is named 'cattle' and is in the same directory as this script.")
        exit()

    # Load Dataset directly from the folder structure
    print("Loading dataset...")

    # Check total files
    total_files = sum([len(files) for r, d, files in os.walk(DATASET_DIR) if any(f.endswith((''.jpg', '.jpeg',
    '.png')) for f in files)])

    if total_files > 5:
        train_ds = tf.keras.utils.image_dataset_from_directory(
            DATASET_DIR, validation_split=0.2, subset="training", seed=123,
            image_size=IMG_SIZE, batch_size=BATCH_SIZE
        )
        val_ds = tf.keras.utils.image_dataset_from_directory(
            DATASET_DIR, validation_split=0.2, subset="validation", seed=123,
            image_size=IMG_SIZE, batch_size=BATCH_SIZE
        )
    else:
        train_ds = tf.keras.utils.image_dataset_from_directory(
            DATASET_DIR, seed=123, image_size=IMG_SIZE, batch_size=BATCH_SIZE
        )
        val_ds = train_ds # Use training set as validation for tiny demo datasets

    TARGET_BREEDS = train_ds.class_names
    print(f"Detected Breeds: {TARGET_BREEDS}")

```

```

# Normalize the datasets (scale pixels to 0-1) for training
normalization_layer = tf.keras.layers.Rescaling(1./255)
train_ds = train_ds.map(lambda x, y: (normalization_layer(x), y))
val_ds = val_ds.map(lambda x, y: (normalization_layer(x), y))

# Initialize the Classifier
classifier = BreedClassifier(num_classes=len(TARGET_BREEDS),
model_weights_path=WEIGHTS_PATH)

# Train the model if weights don't exist
if not os.path.exists(WEIGHTS_PATH):
    print("\nInitiating Training Phase...")
    classifier.train_model(train_ds, val_ds, epochs=10, save_path=WEIGHTS_PATH)
else:
    print("\nPre-trained weights found. Skipping training phase.")

# Inference Testing (Simulating a Field Officer's capture)
print("\n--- Testing Inference ---")
image_handler = ImageHandler()
preprocessor = DataPreprocessor(target_size=IMG_SIZE)

# Safely grab the first valid image from the first breed folder for testing
first_breed_dir = os.path.join(DATASET_DIR, TARGET_BREEDS[0])
valid_extensions = ('.jpg', '.jpeg', '.png')

sample_images = []
if os.path.exists(first_breed_dir):
    sample_images = [f for f in os.listdir(first_breed_dir) if f.lower().endswith(valid_extensions)]

if sample_images:
    test_image_path = os.path.join(first_breed_dir, sample_images[0])

    try:
        raw_image = image_handler.load_image(test_image_path)
        processed_tensor = preprocessor.preprocess(raw_image)

        breed, conf = classifier.predict(processed_tensor, class_names=TARGET_BREEDS)

        record = CensusRecord(breed_name=breed, confidence_level=conf)
        db_status = record.save_to_db()

        print(f"Tested Image: {test_image_path}")
        print("\n--- Output ---")
        print(f"o Detected Breed: {record.breed_name}")
        print(f"o Confidence: {record.confidence_level}%")
        print(f"o Status: {db_status}")
    
```

```

except Exception as e:
    print(f"Inference Error: {e}")
else:
    print(f"No valid test images found in {first_breed_dir} to run the inference test.")

```

13.5 Testing_model.py

```

import os
import cv2
# Import the new CattleDetector along with existing components
from ml_core import (
    ImageHandler,
    DataPreprocessor,
    BreedClassifier,
    CattleDetector,
    get_target_breeds,
    IMG_SIZE,
    WEIGHTS_PATH
)

if __name__ == "__main__":
    # 1. Dynamically read breed names
    TARGET_BREEDS = get_target_breeds()

    if not TARGET_BREEDS:
        print("CRITICAL ERROR: Dataset directory not found or empty.")
        exit(1)

    # 2. Initialize tools (Including the new Detector)
    try:
        print("Initializing models (this may take a moment)...")
        detector = CattleDetector() # Step 1: Cow Detector
        classifier = BreedClassifier(num_classes=len(TARGET_BREEDS),
model_weights_path=WEIGHTS_PATH) # Step 2: Breed Classifier
        preprocessor = DataPreprocessor(target_size=IMG_SIZE)
        image_handler = ImageHandler()
    except Exception as e:
        print(f"Startup Error: {e}")
        exit(1)

    # 3. Prompt for the image path
    print("\n" + "="*50)
    image_path = input("Enter the path to the image: ").strip().strip("\\")

    if not os.path.exists(image_path):

```

```
print(f'Error: File not found at '{image_path}'.')
exit(1)
```

4. Run Two-Step Inference

```
try:
```

```
    # Load the original image
```

```
    raw_image = image_handler.load_image(image_path)
```

```
    print("Step 1: Detecting cattle in image...")
```

```
    # Step A: Use YOLO to find and crop the cow
```

```
    cropped_cow = detector.detect_and_crop(raw_image)
```

```
    if cropped_cow is not None:
```

```
        print("Step 2: Cattle detected! Classifying breed...")
```

```
        # Step B: Preprocess only the cropped section
```

```
        processed_tensor = preprocessor.preprocess(cropped_cow)
```

```
        # Step C: Classify breed
```

```
        breed, conf = classifier.predict(processed_tensor, class_names=TARGET_BREEDS)
```

```
        print("\n--- Output ---")
```

```
        print(f'File:      {image_path}')
```

```
        print(f'Detection:  SUCCESS')
```

```
        print(f'Detected Breed: {breed}')
```

```
        print(f'Confidence:  {conf}%')
```

```
        print("-----\n")
```

```
    else:
```

```
        # If YOLO didn't find class 19 (cow)
```

```
        print("\n--- Output ---")
```

```
        print(f'File:      {image_path}')
```

```
        print(f'Detection:  No Cattle Detected')
```

```
        print("Action:      Breed classification skipped.")
```

```
        print("-----\n")
```

```
except Exception as e:
```

```
    print(f'Inference Error: {e}')
```